

Database System Engineering and Distributed Backend Development

Course Code: 25CS13002E

ROLL NO:2520030001

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Query 1: - CREATE DATABASE

```
2520030001-Aswitha  
CREATE DATABASE IF NOT EXISTS joins_db;  
USE joins_db;
```

Output: -

#	Time	Action	Message	Duration / Fetch
✓ 1	13:55:57	CREATE DATABASE IF NOT EXISTS joins_db	1 row(s) affected	0.031 sec
✓ 2	13:56:01	USE joins_db	0 row(s) affected	0.000 sec

Query 2: -CREATE TABLES

```
CREATE TABLE class (  
    id INT,  
    name VARCHAR(30)  
);  
  
CREATE TABLE class_info (  
    id INT,  
    address VARCHAR(30)  
);
```

Output: -

✓ 3	13:58:04	CREATE TABLE class (id INT, name VARCHAR...	0 row(s) affected	0.125 sec
✓ 4	13:58:07	CREATE TABLE class_info (id INT, address V...	0 row(s) affected	0.047 sec

Query 3: -INSERT VALUES

2520030001-Aswitha

```
INSERT INTO class VALUES
```

```
(1,'Aswitha'),
```

```
(2,'Adhritha'),
```

```
(4,'Amulya');
```

```
INSERT INTO class_info VALUES
```

```
(1,'HYDERABAD'),
```

```
(2,'DELHI'),
```

```
(3,'CHENNAI');
```

Output: -

✓	7	14:01:03	INSERT INTO class VALUES (1,'Aswitha'), (2,'Adhr...	3 row(s) affected	Records: 3 Duplicates: 0 Wamin...	0.031 sec
✓	8	14:01:08	INSERT INTO class_info VALUES (1,'HYDERABA...	3 row(s) affected	Records: 3 Duplicates: 0 Wamin...	0.062 sec

Query 4: - CROSS JOIN

2520030001-Aswitha

```
SELECT *
```

```
FROM class
```

```
CROSS JOIN class_info;
```

Output: -

✓	17	14:04:24	SELECT * FROM class CROSS JOIN class_info L...	9 row(s) returned	0.000 sec / 0.000 sec
---	----	----------	--	-------------------	-----------------------

	id	name	id	address
▶	4	Amulya	1	HYDERABAD
	2	Adhritha	1	HYDERABAD
	1	Aswitha	1	HYDERABAD
	4	Amulya	2	DELHI
	2	Adhritha	2	DELHI
	1	Aswitha	2	DELHI
	4	Amulya	3	CHENNAI
	2	Adhritha	3	CHENNAI
	1	Aswitha	3	CHENNAI

Result 1

Query 5: - INNER JOIN

```

SELECT *
FROM class
INNER JOIN class_info
ON class.id = class_info.id;

```

Output: -

18 14:08:29 SELECT * FROM class INNER JOIN class_info O... 2 row(s) returned 0.000 sec / 0.000 sec

	id	name	id	address
▶	1	Aswitha	1	HYDERABAD
	2	Adhritha	2	DELHI

Query 6: -INNER JOIN

2520030001-Aswitha

```

SELECT class.name,
       class_info.address
FROM class
INNER JOIN class_info
ON class.id=class_info.id;

```

Output: -

19 14:10:36 SELECT class.name, class_info.address FRO... 2 row(s) returned 0.000 sec / 0.000 sec

	name	address
▶	Aswitha	HYDERABAD
	Adhritha	DELHI

Query 7: - NATURAL JOIN

```

SELECT *
FROM class
NATURAL JOIN class_info;

```

Output: -

17	14:10:30	SELECT class.name, class_info.address FROM...	2 row(s) returned	0.000 sec / 0.000 sec
20	14:12:24	SELECT * FROM class NATURAL JOIN class_inf...	2 row(s) returned	0.000 sec / 0.000 sec

	id	name	address
▶	1	Aswitha	HYDERABAD
	2	Adhritha	DELHI

Query 8: - LEFT OUTER JOIN

```
7 • INSERT INTO class VALUES
8   (5, 'Nihitha');
9 • INSERT INTO class_info VALUES
0   (7, 'BENGALURU'),
1   (8, 'PUNE');
```

Output: -

21	14:14:30	INSERT INTO class VALUES (5, 'Nihitha')	1 row(s) affected	0.031 sec
22	14:14:36	INSERT INTO class_info VALUES (7, 'BENGALU...	2 row(s) affected Records: 2 Duplicates: 0 Wami...	0.046 sec

Query 9: - LEFT JOIN

```
SELECT *
FROM class
LEFT OUTER JOIN class_info
ON class.id=class_info.id;
```

Output: -

23	14:16:23	SELECT * FROM class LEFT OUTER JOIN class...	4 row(s) returned	0.000 sec / 0.000 sec
----	----------	--	-------------------	-----------------------

	id	name	id	address
▶	1	Aswitha	1	HYDERABAD
	2	Adhritha	2	DELHI
	4	Amulya	NULL	NULL
	5	Nihitha	NULL	NULL

Query 10: - LEFT JOIN

```

SELECT *
FROM class
LEFT JOIN class_info
ON class.id=class_info.id
WHERE class_info.id IS NULL;

```

Output: -

✓ 24 14:18:46 SELECT * FROM class LEFT JOIN class_info ON... 2 row(s) returned 0.000 sec / 0.000 sec

	id	name	id	address
▶	4	Amulya	NULL	NULL
	5	Nihitha	NULL	NULL

Query 11: - RIGHT JOIN

```

SELECT *
FROM class
RIGHT OUTER JOIN class_info
ON class.id=class_info.id;

```

Output: -

✓ 25 14:20:51 SELECT * FROM class RIGHT OUTER JOIN clas... 5 row(s) returned

	id	name	id	address
▶	1	Aswitha	1	HYDERABAD
	2	Adhritha	2	DELHI
	NULL	NULL	3	CHENNAI
	NULL	NULL	7	BENGALURU
	NULL	NULL	8	PUNE

Query 12: -UNION

2520030001-Aswitha

```
CREATE TABLE first_table(  
  id INT,  
  name VARCHAR(30)  
);  
CREATE TABLE second_table(  
  id INT,  
  name VARCHAR(30)  
);  
INSERT INTO first_table VALUES  
(1,'Akshaya'),  
(2,'Manasvi');  
INSERT INTO second_table VALUES  
(2,'Manasvi'),  
(3,'Ramya');
```

28	14:25:46	CREATE TABLE first_table(id INT, name VARCH...	0 row(s) affected	0.047 sec
29	14:25:50	CREATE TABLE second_table(id INT, name VA...	0 row(s) affected	0.031 sec
30	14:25:54	INSERT INTO first_table VALUES (1,'Akshaya'), (...	2 row(s) affected Records: 2 Duplicates: 0 Wami...	0.046 sec
31	14:25:58	INSERT INTO second_table VALUES (2,'Manasv...	2 row(s) affected Records: 2 Duplicates: 0 Wami...	0.047 sec

```
SELECT * FROM first_table  
UNION  
SELECT * FROM second_table;
```

32	14:27:30	SELECT * FROM first_table UNION SELECT * FR...	3 row(s) returned	0.031 sec / 0.000 sec
----	----------	--	-------------------	-----------------------

	id	name
▶	1	Akshaya
	2	Manasvi
	3	Ramya

Query 13: - UNION ALL

2520030001-Aswitha

```
SELECT * FROM first_table  
UNION ALL  
SELECT * FROM second_table;
```

Output: -

33	14:29:01	SELECT * FROM first_table UNION ALL SELECT ...	4 row(s) returned
----	----------	--	-------------------

	id	name
▶	1	Akshaya
	2	Manasvi
	2	Manasvi
	3	Ramya

Query 14: -INTERSECT (MYSQL DO NOT SUPPORT INTERSECT DIRECTLY)

```
SELECT first_table.*  
FROM first_table  
INNER JOIN second_table  
ON first_table.id = second_table.id;
```

Output: -

34	14:32:12	SELECT first_table.* FROM first_table INNER JOI...	1 row(s) returned
----	----------	--	-------------------

	id	name
▶	2	Manasvi

Query 15: -MINUS (MYSQL DO NOT SUPPORT MINUS DIRECTLY)

```
SELECT first_table.*  
FROM first_table  
LEFT JOIN second_table  
ON first_table.id = second_table.id  
WHERE second_table.id IS NULL;
```

Output: -

35	14:34:55	SELECT first_table.* FROM first_table LEFT JOIN...	1 row(s) returned
----	----------	--	-------------------

	id	name
▶	1	Akshaya

Query 16: -Generate a report showing all students and address availability status.

```

2520030001-Aswitha
SELECT c.id,
       c.name,
       CASE
         WHEN ci.address IS NULL
         THEN 'Address Missing'
         ELSE 'Address Available'
       END AS Status
FROM class c
LEFT JOIN class_info ci
ON c.id=ci.id;

```

Output: -

37

14:37:51

SELECT c.id, c.name, CASE

WH...

4 row(s) returned

0.000 s

	id	name	Status
▶	1	Aswitha	Address Available
	2	Adhritha	Address Available
	4	Amulya	Address Missing
	5	Nihitha	Address Missing

Query 17: -FULL OUTER JOIN (MYSQL DO NOT DIRECTLY SUPPORT FULL OUTER JOIN)

2520030001-Aswitha

```
SELECT *
FROM class
LEFT JOIN class_info
ON class.id = class_info.id
WHERE class_info.id IS NULL
UNION
SELECT *
FROM class
RIGHT JOIN class_info
ON class.id = class_info.id
WHERE class.id IS NULL;
```

Output: -

✓ 38 14:41:25 SELECT * FROM class LEFT JOIN class_info ON... 5 row(s) returned 0.000 sec / 0.000 sec

	id	name	id	address
▶	4	Amulya	NULL	NULL
	5	Nihitha	NULL	NULL
	NULL	NULL	3	CHENNAI
	NULL	NULL	7	BENGALURU
	NULL	NULL	8	PUNE